15 June 1977
SUPERSEDING
MIL-M-38791 (USAF)
1 October 1968

MILITARY SPECIFICATION

MANUALS, TECHNICAL: ENGINEERING INSTALLATION FACILITY STANDARDS (For Ground Communication-Electronic and Meteorological Equipment (C-E-M)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers general content requirements for preparation of technical manuals that will support pre-engineering, site adaptation, planning, installation, and testing of a C-E-M facility in varied environments. The term "facility" used herein refers to a grouping or configuration of two or more ground C-E-M equipments, designed to satisfy a specific operational requirement. Technical manuals prepared in accordance with this specification will contain technical information applicable to specific facilities. The manuals will be used to engineer the installation of complex C-E-M facilities and to test such facilities, after installation, for proper adjustment and operation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-P-38790	Printing Production of Technical Manuals: General Requirements For

PUBLICATIONS

Air Force

AFM 100-18	USAF Ground Communications Electronics Program Management

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Coverage. The installation-engineering facility manual is concerned with equipment compatibility, equipment interface and cabling interconnections between equipments that allow the facility to function as an operational entity. Emphasis shall be placed on equipment configuration, unified installation, floor plans, engineering planning criteria, installation criteria, and facility tests.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC/LOLMP, WPAFB OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

TMSS

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3.1.1 Preparation. The general requirements for preparation of text and artwork shall be in accordance with MIL-M-38784 and MIL-M-38790.

3.2 Arrangement. Manuals shall contain the following elements in the order indicated:

Title Page
List of Effective Pages
Table of Contents
List of Illustrations
List of Tables
Introduction
Chapter 1 - General Information
Chapter 2 - Engineering
Chapter 3 - Installation
Chapter 4 - Test
Appendix A, List of Reference Publications
Alphabetical Index

3.2.1 Title page and list of effective pages. These shall be prepared in accordance with the requirements of MIL-M-38784. In addition, the following identification shall appear on the title page, two lines below and in similar type as the term "Technical Manual".

AIR FORCE COMMUNICATIONS SERVICE
(C-E STANDARD)

3.2.1.1 Facility title. The title page shall identify the facility by means of the exact title and C-E facility code number as indicated in AFM 100-18, Atch 3. The specific configuration shall be indicated as a secondary title under the facility code and title.

3.2.2 Table of contents, list of illustrations and list of tables. These shall be prepared in accordance with the requirement of MIL-M-38784.

3.2.3 Introduction. The introduction shall identify the facility by its code and descriptive title and shall also identify major and minor equipments which compose the specific configuration to be described. The introduction shall include a brief statement of the manual's purpose and use, and outline briefly the contents of each chapter. If possible, a photograph of a standard configuration layout of the facility shall be used as a frontispiece. When a photograph is not possible because of the size, or the physical area required by the facility, an artist's sketch or diagram shall be presented. In either case, appropriate call-outs of equipment, structures, and special physical consideration shall be made.

3.2.4 Chapters. Chapters shall not normally be composed of sections unless the complexity and length of the data contents require section breakout. Normally, chapters may be broken into primary and subordinate paragraphs. Tabular material should be used whenever possible to present adequate information coverage. If the information required for any chapter is not applicable because of special facility design or application, the chapter title and number shall be shown on a page with the statement NOT APPLICABLE inserted immediately below the title; the next applicable chapter shall start, with normal spacing, immediately thereafter. Each chapter shall be started on a right hand page.

3.2.4.1 Sections. When contents of a chapter are broken into sections, each section shall completely cover the subject heading of the section. If section format is used and a particular section does not apply because of facility characteristics, the section number and title shall be shown on a page with the statement NOT APPLICABLE inserted immediately below the title; the next applicable section shall start, with normal spacing, immediately thereafter. A section may start on either a left or right hand page.

3.2.5 Chapter 1, General Information. This chapter shall describe the facility, state its purpose in terms of its function and shall, by means of functional descriptions of the facility and its components, describe how the function is accomplished. This chapter shall also identify the interfaces of the facility with other portions of a system, sub-system, or facilities with reference to the illustrations specified in 3.2.5.2.

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3.2.5.1 Description and purpose. A brief but thorough summary of the facility shall be given in terms of its physical description. The facility's purpose shall be described in terms of its function and operational use. The relationship of the facility to interconnecting facilities shall be explained in terms of its major equipment function. A short summary of how the function of the facility is accomplished shall be included as a briefing orientation to engineering and installation personnel.

3.2.5.2 Illustrations. In addition to the frontispiece (3.2.3), block diagrams and other type diagrams shall be presented to show functional relationships between equipments, direction of signal flow, and interconnections with other facilities, remote control, or monitoring devices.

3.2.5.3 Tables. Tables are required to provide field engineers with adequate data for pre-planning purposes. These tables consist of:

Leading Particulars
Capabilities and Limitations
Equipment Supplied
Equipment Required but not Supplied

3.2.5.3.1 Leading particulars. Tables shall list the logistic characteristics of the facility and the maximum-minimum variations from a standard equipment configuration. Tables shall present required facility floor space; dimensions and weight of individual equipments plus floor and wall projections required for inside plant installations; electromagnetic compatibility aspects; such as, transmitter power, receiver sensitivity and bandwidth of equipments within the facility configuration. Tables shall include input power requirements, both AC and DC, including kVA loading (connected and demand); frequency, power factor, and number of phases for AC requirements; allowable frequency variation and voltage levels with allowable variations, plus other electrical data that will enable an engineer to plan an installation. Identification of all cables used in the facility shall be presented. Absolute identification shall be made by nomenclature, symbol, number, function, cable type, number and size of conductors, length, and terminal connectors. This presentation is required for all inter-equipment cabling and, when applicable, for input and output cables that link the facility to another or to auxiliary equipments. Special logistic techniques and procedures for transportation and storage shall also be listed in the tables.

3.2.5.3.2 Capabilities and limitations. The functional characteristics of the facility shall be identified in terms of types of operation, power output, frequency, pulse characteristics, antenna characteristics, RFI malfunctioning levels, sensitivity, selectivity, storage capacity, and memory type viewing parameters. The capabilities of the facility shall be further identified in terms of maximum range, coverage resolution and accuracy. In addition, limiting factors of the facility shall be listed in terms of climatic effects; such as, ambient and operating temperatures, pressure, humidity, wind and ice loading; abnormal power stability requirements, maximum tolerances in frequency and voltage deviations; excessive floor or tower loading; and abnormal clearance requirements for installation, operation, and maintenance.

3.2.5.3.3 Equipment supplied. This table shall contain an itemized list of all equipments and major components that are part of the facility. When engineered SFELs (Standard Facility Equipment Lists) exist, the equipments should be listed under the specific SFEL package. Identification shall be made by military nomenclature when it exists, or by commercial nomenclature, by common name, and by manufacturer's part number. A very brief descriptive statement covering the technical purpose of each identified item shall also be presented.

3.2.5.3.4 Equipment required but not supplied. This table shall list all equipment required to complete the facility configuration and SE (Support Equipment) required for installation and test that are not supplied through facility procurement. The listed equipment shall be adequately identified.

3.2.6 Chapter 2, Engineering. This chapter shall provide sufficient data to enable a field engineer to pre-engineer, site adapt and accomplish engineering planning for installation of a facility to meet a functional requirement in any specific geographical area. Data shall be presented in the following order:

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Siting Considerations
Supporting Structures
Power Requirements
Layout Planning

3.2.6.1 Siting considerations. Basic information required to site adapt a standard facility configuration shall be presented. It shall consist of three primary areas of consideration: electronic, logistic, and geographical.

3.2.6.1.1 Electronic. Criteria shall be given for clear area requirements for radio frequency propagation, radio frequency interference shielding and filtering, reflections, random noise sources, need for siting away from shock sources; such as, blasting, heavy air or ground traffic, nuclear explosion. In addition, the electronic criteria shall take into account the possibility that a standard configuration may be augmented, or that specific equipment may be deleted under certain operating conditions.

3.2.6.1.2 Logistic. The siting criteria for logistic support shall contain information that will identify type and quantity of support required for physical installation, maintenance and operation of the facility from electrical power sources, commercial communications facilities, fuel sources, water supply sources, drainage/sanitary waste systems, access roads and transportation services. The area required for storage and assembly of facility components during site preparation, installation, and follow-on operation and maintenance, shall also be described and diagrammed.

3.2.6.1.3 Geography. The geographical siting criteria shall contain a description and diagram for a normal or average site. The description shall include: dimensions and area of total site, including clear zones (areas); soil bearing characteristics required for installation and subsequent operation; grading and drainage limits; building and equipment electrical grounding. Limiting climatic and natural conditions that affect site construction, installation, and operation shall be listed and described.

3.2.6.2 Supporting structures. Complete structure data is required for any proposed structure or portion thereof. The information shall include novel supporting structures or the criteria necessary to select or redesign interiors of existing structures to house facility elements. When new buildings are required, they shall be designated as permanent or semi-permanent. Type of building should be classified as fireproof, semi-fireproof, or ordinary. These requirements shall be indicated for both inside and outside plant.

3.2.6.2.1 Inside plant. The description of inside plant shall include, but not necessarily be limited to, the following criteria:

a. A layout plan showing and describing gross floor space requirements for the facility, including access, maintenance, storage, operating and special use areas, minimum access door clearance (height and width required for access and egress of largest non-reducible facility component).

b Required clearance around equipment in operating area and other facility related spaces.

c Required floor to clear ceiling heights.

d. Design live and wind loads with particular emphasis to unusual live load requirements.

e. Special architectural features; such as, cable openings, cable trenches, raised floor, acoustic treatment, outsized doors, removable wall panels.

f. Special structural features; such as, elevators, cranes, hoists, concentrated floor loads, cable anchors, structural ferrous bonding.

g. Technical equipment heat gain per facility element.

h. Inside summer and winter design dry bulb temperature, relative humidity.

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i. Requirement for heating, air conditioning and mechanical ventilation, humidity control, plumbing and special mechanical features.

j. A statement of lighting requirements, including general and special lighting techniques and intensities, and hazard lighting.

k. A qualitative and quantitative description of the electric power required for operation and maintenance of the facility technical equipment, including identification and number of circuits, plus a riser diagram. Where the facility demands uninterrupted power, as a condition of specified operation, the elements so affected shall be identified in the breakout of power consumption. Facility kVA power consumption shall be given in both demand and connected load.

l. Facility equipment, structures grounding, and lightning protection plan for both inside and outside plant equipment.

m. Specific requirements for facility related intercom, telephone, and timing systems.

n. Location and dimension of RF shielded and classified communications equipment areas.

o. Orientation of inside plant to outside plant.

p. Any novel feature peculiar to the facility.

3.2.6.2.2 Outside plant. The following points shall be explained and shown in the construction design criteria:

a. Antenna site plot plan, including radomes, antenna warning lights, shelters, waveguide and transmission line routes, guys and guy anchors, grounding and fencing, etc. Controlling dimensions of equipment and structure spaces and clearance areas shall be given.

b. Antenna/antenna component mounting plate(s) dimensional details.

c. Antenna support foundation(s) and guy anchor reaction data.

d. Description and physical data relating to transmission lines and cables and their suspension structures.

e. Description and location for underground or buried cable, trenching, ducting, direct-burial cable, manholes, handholes or pull boxes for signal or control cables.

f. Location and description of access roads, parking areas, equipment pads, and similar features required for installation, operation, and maintenance of the facility.

3.2.6.3. Power requirements. Power requirements shall be described which will indicate, in addition to that previously listed, interface between electrical construction effort and technical facility installation responsibilities.

3.2.6.3.1 Electrical system data. The number, electrical rating, and location of switching and distribution panels shall be given. As a part of this data, a listing of circuit breaker assignments for the technical equipment installation/operation/maintenance shall be included. The individual branch circuit breaker is to be described by its panel location and rating for frequency, voltage, phase, capacity, and whether automatic closure type is required.

3.2.6.4 Layout planning. Although the information required in 3.2.6.1 through 3.2.6.3 will provide engineering information requisite for installation planning, the information required herein shall correlate the data into a brief description of the principal factors that influence arrangement of the equipments, components, and hardware which make up the facility. These considerations shall entail such information as:

a. Electronic: electromagnetic interference and potential radiation effects.

b. Functional association of equipments.

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c. Allowable or design lengths of transmission lines and cables.

d. Critical lengths of interconnecting signal cables.

e. Circuit terminations, including tie-in with other facilities.

f. Remote control and monitoring (microwave versus land line or other type of transmission).

g. Special intra- and inter-communications.

h. Off-site location of commercial power and communication facilities.

i. Physical consideration such as:

1. Location of installed facilities (heating, air conditioning, and so forth).

2. Efficiency of total installation, including ease of operation and maintenance (access requirements for operation, maintenance, installation, and removal).

3. Space provision for potential expansion.

4. Security fencing, parking area requirements.

5. Required intensity of room illumination (foot-candles).

6. Allowable intensity of sound in decibels.

7. Topography.

8. Water table limitations when underground structures are used.

9. Permissible live loading and snow loading for flat roofs.

10. Economy of construction and installation.

j. Tabulation of minimum-maximum or, when applicable, exact distances between:

1. Equipments, associated control units and indicators, electrical and physical plant facilities, and major equipment components.

2. Equipments and building walls and ceilings.

3. Similar and non-similar antennas and associated equipments.

4. Antennas and buildings.

5. Power and signal cables.

6. Antennas, and power and signal cables.

7. Similar and non-similar transmission lines (power and communications).

3.2.6.5 Installation-Engineering drawings. Chapter 2 shall contain reference to all drawings required for engineering the installation of the facility. References shall be to dimensional drawings concerned with a typical or standard layout, including an overall site plan, equipment arrangement, simplified wiring diagrams, simplified floor and plot plans showing cross section elevations and arrangement of utilities. Additionally, the typical configuration shall be supplemented with minimum and maximum equipment configuration if variations are applicable.

3.2.7 Chapter 3, Installation. This chapter shall contain the information and instructions required to install a specific facility. Data content shall include, but not be limited to, the following items:

Equipment and materials
Installation instructions
Manpower requirements

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3.2.7.1 Equipment and materials. Complete identification shall be given of all equipments and materials that will be required to accomplish an installation. Such information shall normally be presented in tabular form and shall consist of:

a. A list of all engineered and approved SFELs that are applicable to the facility.

b. A listing of equipment and material required but which are not included in approved SFELs. This listing shall include nomenclature (military when applicable, commercial if no military nomenclature exists) plus federal stock numbers of each listed item.

c. A listing of all tools, test equipment and material handling equipment that are required to install and test the facility. The identification shall include nomenclature, federal stock number, and brief description of each item.

3.2.7.2 Installation instructions. Complete instructions shall be presented that will enable installation personnel to install the facility in accordance with established practices. These instructions shall not repeat contents of equipment installation unless such information is required for clarity. Standard Air Force construction and installation practices and procedures shall not be presented, however, installation practices and procedures that are peculiar to the installation of the facility shall be presented. Required data shall include, but not be limited to:

a. Instructions in the use of special installation tools and material handling equipment. Illustrations shall be used as required.

b. Instructions and requirements, if applicable, for sequential installation of equipments, components and materials that make up the facility.

c. Installation instructions for:

1. Inside plant equipment and material, including interconnecting cabling and wiring,

2. Erection of self-supporting and guyed towers, radomes, antennas and other related housing and supporting structures,

3. Tower mounted and other outside plant equipment, including use of special lift equipment,

4. Construction of overhead, underground and buried power, control and communication lines,

5. Fabrication of special signal cables that are not included in SFELs.

6. Waveguides and signal transmission lines.

3.2.7.2.1 Installation drawings. Chapter 3 shall contain reference to all drawings required to install the facility. The information shall contain a list of standard building or room construction drawings, plus the applicable specifications that govern such construction drawings, plus the applicable specifications that govern such construction. A list shall also be included, when applicable, of drawings that pertain to outside plant; such as, tower construction and erection, plus the applicable specifications as these drawings pertain to facility installation. In addition to lists of Air Force drawings that apply to the facility, this chapter shall contain simplified drawings, wiring diagrams, wiring and cable lists (drawings), and miscellaneous illustrations that are necessary to supplement existing data, or to clarify installation instructions presented in the standard.

3.2.7.3 Manpower requirements. Total manpower requirements to install the facility shall be presented. This information shall be expressed in man-days or man-weeks and shall include installation plus time required to test the installed facility. The information shall show work categories and the amount of time required to accomplish each category.

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3.2.8 Chapter 4, Test. This chapter shall contain procedures for conducting tests required to insure that the installed facility is technically adequate to accomplish its operational objective. The information shall include, but not be necessarily limited to, these types of tests: Pre-shakedown, shakedown and operational.

3.2.8.1 Pre-shakedown. The pre-shakedown test shall include the requirement for physical inspection to verify conformance with installation requirements and safety; electrical measurements to confirm the adequacy and stability of primary power, voltage, frequency, etc; and preliminary adjustments to be made prior to shakedown and operational tests.

3.2.8.2 Shakedown. The shakedown test is essentially a running time test for each electronic part that composes the facility. The manual shall specify the duration of time required for the shakedown test in order to insure that the facility will operate adequately to accomplish its operational requirement. Real or simulated operating conditions shall be prescribed so that the whole facility and its parts are at normal functioning voltage, current, temperature, and so forth.

3.2.8.3 Operational. A description of the tests required to assure that all the operational functions of the facility have been met shall be given. The purpose of these tests is to provide procedures for accurate determination of performance levels. The tests shall include detailed instructions, including required duration of tests, and shall make provision for permanent recording of data. Flight and electronic countermeasures (ECM) tests shall be included, when applicable, in the list of tests to be conducted, however, the requirement for flight tests shall be held to an absolute minimum, relative to accurate determination of facility performance levels.

3.2.8 3.1 Test sequence. The preferred sequence of tests shall be stated. In sequence listing, a step-by-step procedure for each test shall be presented, or a reference shall be made to the publication in which the prescribed procedures can be found Test procedures shall include:

a. Performance specifications.

b. Test interface between test equipment and the facility, to include cables, plugs, attenuators, etc.

c. Test equipment required.

d. Sample test data sheets, or the establishment of the requirement for oscilloscope photographs, pen charts, computer read-outs, and such data.

e. Test configuration diagram.

NOTE: The following statement shall appear following the description of operational tests:

"Accomplish the performance, alignment, and adjustment routines on the installed facility in the presence of representatives of the using command."

3.2.9 Appendix A, List of Reference Publications. Inasmuch as the facility manual will not duplicate information contained in other publications, a list of reference publications shall be made up as an appendix to the manual. The list shall identify applicable reference publications by number and title identifier. It shall include publications and documents to which reference is made in the text plus other publications information presented in the manual. The list shall also include technical manuals pertinent to the individual equipments and materials that comprise the facility in all variable configurations and standard installation practices technical orders (SIPTOs) prescribed for Air Force use.

3.2.10 Alphabetical index. An alphabetical index is required in all manuals that contain 100 or more pages. The index shall list in alphabetical order the main topics discussed in the text. Location of listed topics shall be identified by paragraph number.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The requirements of MIL-M-38784 and MIL-M-38790 are applicable.

5. PREPARATION FOR DELIVERY

5.1 The requirements of MIL-M-38784 and MIL-M-38790 are applicable.

6. NOTES

6.1 Intended use. The engineering installation facility manuals prepared in accordance with the requirements of this specification are intended for use by experienced Air Force personnel trained and familiar with the installation-engineering and installation of similar type facilities.

6.2 Definitions.

6.2.1 C-E-M equipment. The physical components of a facility, usually denoting a major end item having a military nomenclature (AN/-, TS-, OA-, etc) as opposed to a part number.

6.2.2 C-E-M facility. A group of equipments or components engineered and installed to perform an operational function in support of a requirement, system, or mission.

6.2.3 Site. A physical location of ground C-E-M equipments or facilities, such as: Station, Base, Depot, ALC, Field, Fort, Launching Area, Camp, Installation, and Arsenal.

6.2.4 System. A numbered part of a Mission Category, such as: 480L, 465L, etc, that may consist of one or more facilities or sites.

6.2.5 SFEL. Standard Facility Equipment List.

6.3 Review activities. In addition to the review activities listed below, HQ Air Force Communications Service (AFCS/1842 EEG/EEISD, Richards-Gebaur AFB Missouri 64030) shall review drafts of this specification.

Custodian:
Air Force - 16

Preparing Activity:
Air Force - 16

Review Activities:
Air Force - 01, 10, 99

Project TMSS-P439

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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